Disclaimer

This document is an exam summary and follows the given material of the lecture *Bioelectronics*. Its contribution is a short summary that contains the most important concepts, formulas and algorithms. Due to curriculum content updates, some content may not be relevant to future versions of the course.

I do not guarantee the accuracy or completeness, nor is this document endorsed by the instructors. Any errors that are pointed out to me are welcome. The complete LaTeX source code can be found at `https://github.com/tstreule/eth-cheat-sheets`.

Bioelectronics

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General

Formeln

Circle: $U_{\text{circle}} = 2\pi r$, $A_{\text{circle}} = \pi r^2$

Constants:

$h = 6.626\text{E}{-34}\,\text{Js} = 4.135\text{E}{-15}\,\text{eVs}$, $\hbar = \frac{h}{2\pi}$, $hc = 1.986\text{E}{-25}\,\text{Jm}$

$\epsilon_0 = 8.85\text{E}{-5}\,{}^{\text{As}}/_{\text{Vm}}$

$\mu_0 = 4\pi\text{E}{-7}\,{}^{\text{N}}/_{\text{A}^2}$

$k_B = 1.38\text{E}{-23}\,{}^{\text{J}}/_{\text{K}} = 8.617\text{E}{-5}\,{}^{\text{eV}}/_{\text{K}}$

$q = 1.602\text{E}{-19}\,\text{C}$, $m_e = 9.109\text{E}{-31}\,\text{kg}$, $m_p = 1.672\text{E}{-27}\,\text{kg}$

$F = 96485\,{}^{\text{C}}/_{\text{mol}}$ (Faraday)

$R = N_A k_B = 8.314\,{}^{\text{J}}/_{\text{mol K}}$ (Ideal gas constant)

$N_A = 6.022\text{E}23\,{}^{\text{particles}}/_{\text{mol}}$ $\quad 0\,°\text{C} = 273.15\,\text{K}$

Einheiten

Druck	$1\,\text{Pa} = 1\,{}^{\text{N}}/_{\text{m}^2} = 1\,{}^{\text{J}}/_{\text{m}^3} = 10\,{}^{\text{g}}/_{\text{cm}\cdot\text{s}^2}$
Induktivität	$1\,\text{H} = 1\,{}^{\text{Vs}}/_{\text{A}} = 1\,\Omega\text{s}$
Power	$1\,\text{W} = 1\,{}^{\text{J}}/_{\text{s}} = 1\,\text{VA}$
electron volt	$1\,\text{eV} = 1.602\text{E}{-19}\,\text{J} = 23.06\,{}^{\text{kcal}}/_{\text{mol}}$
Charge	$1\,\text{C} = 1\,\text{As}$
Energy	$1\,\text{J} = 1\,{}^{\text{kg m}^2}/_{\text{s}^2} = 1\,\text{Nm} = 1\,\text{VAs} = 1\,\text{CV} = 1\,\text{Ws}$

Good to know

Power in dB	$10\log_{10}\frac{I}{I_0}$
Intensity	$I = \frac{\text{avg. Power } (P)}{\text{area } (A)}\left[\frac{\text{W}}{\text{m}^2}\right]$
avg. Power	$P = \frac{\text{avg. Work in cycle } (\overline{W})}{\text{cycle } (T)}\left[\text{W} = \frac{\text{kg m}^2}{\text{s}^3}\right]$
(avg.) Work	$\overline{W} = \int_{\text{cycle}} \frac{1}{\text{cycle}(T)} \cdot \vec{F}\cdot\vec{x}\,\text{d}t\,[\text{J} = \text{Ws}]$

mass m vibrates with an amplitude a along x-axis:

$\vec{x}(t) = a\sin(\omega t)\cdot\vec{e}_x$, $\quad \omega = 2\pi f = 2\pi/T$, $\quad \vec{F}(t) = m\ddot{\vec{x}}(t)$

NuS

Induktivität	$u(t) = L\frac{\text{d}i_L}{\text{d}t} \;\circ\!\!-\!\!\bullet\; U(s) = sL\,I_L(s)$
Konduktivität	$i(t) = C\frac{\text{d}u_C}{\text{d}t} \;\circ\!\!-\!\!\bullet\; U(s) = \frac{1}{sC}U_C(s)$
Transformator	$u_1 = L_1\frac{\text{d}i_1}{\text{d}t} - M\frac{\text{d}i_2}{\text{d}t}$, M: mutual inductance
LCR-Schwingkreis	$\omega_0 = 2\pi f_0 = 1/\sqrt{LC}$

Laplace

$\lambda\,u(t) + \mu\,v(t)$	$\lambda\,U(s) + \mu\,V(s)$
$u(at)$, $a>0$	$\frac{1}{a}\,U(\frac{s}{a})$
$u(t-t_0)$	$\text{e}^{-st_0}\,U(s)$
$\text{e}^{-at}\,u(t)$	$U(s+a)$
$(-t)^n u(t)$	$U^{(n)}(s)$
$u^{(n)}(t)$	$s^n U(s) - \ldots - u^{(n-1)}(0)$

t^n	$\frac{n!}{s^{n+1}}$		$t^n\text{e}^{-at}$		$\frac{n!}{(s+a)^{n+1}}$
e^{-at}	$\frac{1}{s+a}$		$\sin(at)$		$\frac{a}{s^2+a^2}$
te^{-at}	$\frac{1}{(s+a)^2}$		$\cos(at)$		$\frac{s}{s^2+a^2}$

1 Biosensors

1.1 Label assay vs. label-free

Label assay (sandwich): 

Label-free assay (OWLS): 

non-specific binding: 

other interactions that could happen, except the angle binding.

1.2 Rules of Chemistry and Physics

Bracket notation indicates concentration A + B $\underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}}$ AR

equilibrium constant $K = \frac{k_{-1}}{k_1} = \frac{[A][R]}{[AR]}$

affinity constant $K_a = 1/K$

cont. flowing cell $[AR] \sim q * \text{signal}$, $q = const$

$[R] \sim R_0 = \text{initial receptor density}$

1.3 Sensitivity and Specificity

Sensitivity: true positive rate (% of correctly identified +)

Specificity: true negative rate (% of correctly identified −)

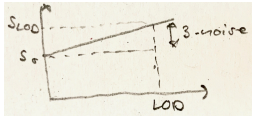
Compensate Sensitivity through LOD.

1.3.1 Limit of Detection (LOD) (Sensitivity)

Limitation by non-specific binding (NSB) → noise at zero analyte

LOD $\quad LOD = 3\cdot\text{noise}/\frac{\text{d}S}{\text{d}[A]}$

$S_{LOD} = S_0 + 3\cdot\text{noise}$



LOD for intensity/signal $\quad LOD_{NSB} = \langle I_{NSB}\rangle + 3\,\sigma(I_{NSB})$

Lowest detectable intensity $\quad \frac{\langle I_{POI}\rangle}{LOD_{NSB}} = 1$ (POI: proteins of interest)

Detectable #proteins $\quad N\# = \frac{\Gamma}{m_{\text{protein}}}$ $\quad [\Gamma] = {}^{\text{pg}}/_{\text{mm}^2}$ detection limit

2 Optical Microscopy

2.1 Reflection and Refraction

Law of Reflection $\quad \theta_{\text{inc}} = \theta_2$

Law of Refraction $\quad \frac{n_{\text{inc}}}{n_2} = \frac{\lambda_0/\lambda_{\text{inc}}}{\lambda_0/\lambda_2} = \frac{\sin\theta_2}{\sin\theta_{\text{inc}}}$ $\quad$ for $\theta_{\text{inc}} < \theta_c$

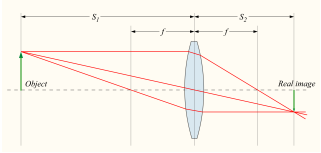
Total reflection $\quad \sin\theta_c = n_2/n_{\text{inc}}$ $\quad$ **always total** if $n_{\text{inc}} < n_2$

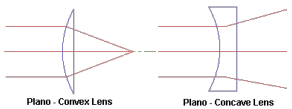
Paraxial approx. $\quad \theta \simeq \sin\theta \simeq \tan\theta$, $\quad \theta \ll 1$

Thin lens approx. $\quad R \ll S_o, S_i$

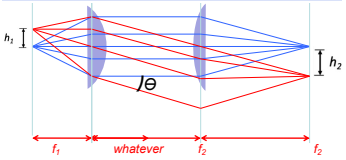
Lens makers formula $\quad \frac{1}{S_o} + \frac{1}{S_i} = \frac{1}{f} = \frac{n_{\text{lens}} - n}{n}\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

2.2 Ray Tracing



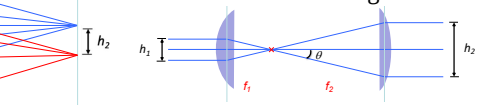


2.2.1 Simple Microscope



2.2.2 Beam Expander

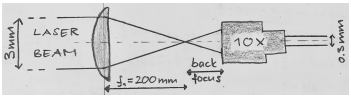
for light source



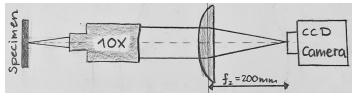
Magnification $\quad M = h_2/h_1 \overset{*}{=} f_2/f_1$ $\quad$ * since $h_1 = f_1\sin\theta$

2.2.3 Simple "modern" Microscope

(uniform!) Illumination: Camera: ($M = f_{tube}/f_{obj}$)



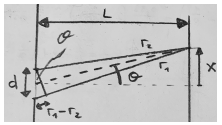
(Objective lens)



(Tube lens)

2.3 Fundamental Limits of Lenses & Resulution

Due to a **finite aperture** will points presented as the PSF (point-spread).



First zero occurs at: $\quad \theta \overset{(\theta\ll1)}{\simeq} \sin\theta \simeq 1.22\frac{\lambda_0}{d}$

d: radius of the aperture

- $\sin\theta \simeq \tan\theta = x/L$
- $r_1 - r_2 = d\cdot x/L$

Rayleigh criterion maximum of the one is on the minimum of the other curve (FWHM)

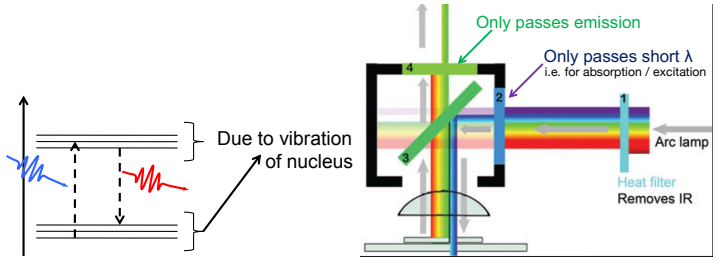
Resolution $\quad \Delta r \simeq f\sin\theta \times M \simeq 1.22\frac{\lambda_0 f}{D} \times M = 0.61\frac{\lambda_0 n}{\text{NA}} \times M$

Numerical Aperture $\quad \text{NA} = n\sin\theta \simeq n\frac{D}{2f}$ $\quad n$: refractive index,

D: aperture size of lens, f: focal length of lens

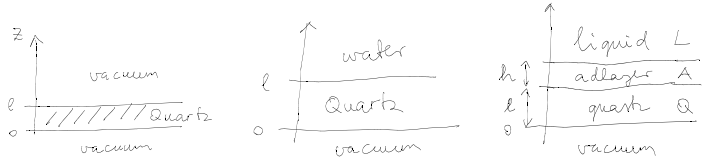
2.4 Fluorescence Microscopy

Emitted photon has **less energy**. → Absorption ≠ emission spectr.



3 Mechanical Sensors

3.1 Viscoelastic media



Remark: Calculate *always* at first in ${}^{\text{rad}}/_{\text{s}} \xrightarrow{\times 1/2\pi}$ Hz.

3.1.1 Crystal with thickness l

Resonance frequency $\omega_n = n \cdot \omega_0 = n \cdot \frac{\pi}{l}\sqrt{\frac{\mu_Q}{\rho_Q}}$ [rad/s]

char. wavelength $\lambda = 2l/n$

3.1.2 Elastic plate in water

Resonance frequency $\omega_n = n\omega_0 + \Delta\omega_n$

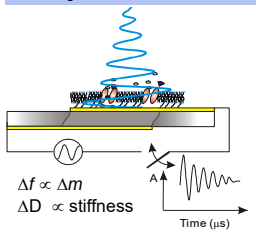
frequency shift $\Delta\omega_n = -\sqrt{n}\sqrt{\frac{\rho_L \eta_L \omega_0}{2}}\frac{1}{l\rho_Q}$, η_L: viscosity

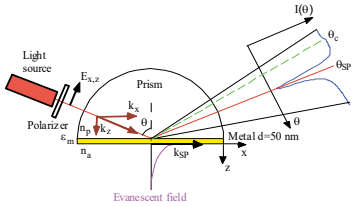
3.1.3 Elastic plate in water with adlayer

In fact is the *resonance frequency* dependent on the adlayer height. But if it is "enough" small, the Sauerbrey approx. is sufficient.

Sauerbrey eq. $\Delta f_n = 2\pi\omega_n = -n\frac{\Delta m}{C}$ $C = \frac{\sqrt{\rho_Q \mu_Q}}{2f_0^2} = 17.7 \frac{\text{ng}}{\text{cm}^2\text{Hz}}$

3.2 QCM-D *vs.* SPR techniques





$\Delta m = -\frac{C}{n}\Delta f_n$

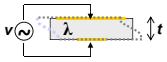
$\Delta m = d\frac{n_{\text{protein}} - n_{\text{buffer}}}{\text{d}n/\text{d}c}$

$\text{LOD}_{\text{QCM}} = C \cdot \delta f \simeq 1\,\text{ng/cm}^2$ $\text{LOD}_{\text{SPR}} = C \cdot \delta\theta \simeq 0.1\,\text{ng/cm}^2$

3.3 QCM: Quartz Crystal Microbalance

Resonance cond. $f = \frac{n\,v}{\lambda} = \frac{n\,v}{2t}$

Dissipation $D = \frac{1}{\pi f \tau}$, τ: time until $\frac{U_{\max}}{\text{e}}$



Modeling of the QCM-D response in air / aqueous solution:



G = G' + jG"
= μ + j2$\pi f\eta$

η: viscosity ($= \frac{G''}{\omega}$), μ: elasticity ($= G'$),
ρ: density, d: thickness

3.4 Strain Gauge

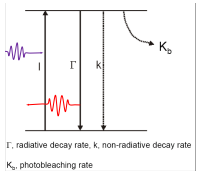
Resistive strain g. $\frac{\Delta R}{R} = k\frac{\Delta l}{l} = k\epsilon$, ϵ: Strain, k: Gauge factor

Capacitive strain g. $C = \epsilon_0\epsilon_r\frac{A}{d}$, (displacement: $d \to d + \Delta d$)

4 Fluorescent Probes

4.1 Fluorescence Statistics

Photobleaching: A fluorescent molecule can emit a limited #photons by excitation before it irreversibly converts to a non-fluorescent molecule.



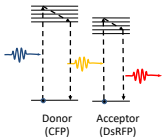
Quant. Yield $Q = \frac{\Gamma}{\Gamma + k + K_b} \sim 0 - 98\%$ (Efficiency)

Fluorescence **Lifetime** $\tau = \frac{1}{\Gamma + k + K_b} \sim 1\,\text{ns}$

Fluorescence emission is a statistical process that is characterized by exponential decays.

Molecules in *excited* state $\frac{\text{d}N_e}{\text{d}t} = -\left(\frac{1}{\tau_1} + \ldots + \frac{1}{\tau_n}\right)N_e$

4.2 FRET – Fluorescence Resonance Energy Transfer



Seeing a different $\lambda_{\text{emission}}$ tells us that two molecules are close.

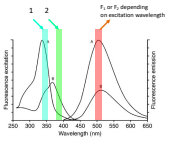
Efficiency $E = \frac{R_0^2}{R_0^2 + r^6} = 1 - \frac{\tau_{\text{DA}}}{\tau_{\text{D}}} = 1 - \frac{I_{\text{DA}}}{I_{\text{D}}}$

where $\tau_{\text{D(A)}}, I_{\text{D(A)}}$ Lifetime/Intensity of donor emission

on in the absence/presence of acceptor.

4.3 Calcium Imaging → too slow for t dependency meas.

Calcium ion cannot be visualized/tagged directly: → Design molecules with optical properties that change upon calcium binding.



Single & dual wavelength measurements:

Concentration $[\text{Ca}^{2+}]_i = K_{\text{d,eff}}\frac{R - R_{\min}}{R_{\max} - R}$ where

for single $R \equiv F$ and $K_{\text{d,eff}} = \frac{[\text{Ca}^{2+}]_i \times (F_{\max} - F)}{F - F_{\min}}$

for dual. $R = F_1/F_2$ (F: fluorescence)

The binding of Ca^{2+} leads to...

- change in fluorescence **intensity** but not wavelength change
- a **shift** in excitation (and/or emission) peaks ("dual wavelength")
- changes in fluor. resonance energy transfer (**FRET**) & life time

4.3.1 Delivery of Calcium Indicators

Loading cells Once the molecule got cleaved (spalten), it cannot go out and gets fluorescent.

Introduce **fluoresc. proteins** or (natural) **FRET proteins** Proteins change emission rate when Ca binds.

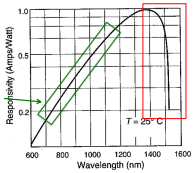
Relative change ($\Delta F/F$) in fluorescence emission of EGFP can then be meas. directly.

Note: The dye responds *fast*, but it is *slow* to recover. Ca dissociation rate (i.e. recovery) depends on dye's affinity for Ca.

5 Detection and Noise

5.1 Signal

For statistically independent, "uncorrelated", events we use



Poisson $P_{\overline{n}}(n) = \frac{\overline{n}^n}{n!}\text{e}^{-\overline{n}}$ $\overline{n}$: mean $\sigma_n = \sqrt{\overline{n}}$: var

Signal current $\langle I_{\text{sig}}\rangle = \eta q \overline{n}/\Delta t$

Responsivity $\langle I_{\text{sig}}\rangle/P = \eta q\lambda/(hc)$

Cutoff λ $\lambda(\mu\text{m}) = 1.24/E(\text{eV})$

5.1.1 Fundamental Noise Sources

Noise

shot $N_{\text{s}} = 2R\eta q B\langle I_{\text{sig}}\rangle = \langle I_{\text{shot}}^2\rangle R$, $2B \simeq 1/\Delta t$

dark $N_{\text{d}} = 2R\eta q B\langle I_{\text{dark}}\rangle$

thermal $N_{\text{J}} = \langle I_{\text{johnson}}^2\rangle R = \sqrt{\frac{k_{\text{B}}TB}{R}}^2 R$

readout $N_{\text{r}} = N_{\text{J}} + N_{\text{amplifier}}$

$N_{\text{tot}} = N_{\text{s}} + N_{\text{d}} + N_{\text{r}}$ $\sigma_{\text{tot}}^2 = \sum\sigma_i^2$

SNR $\text{SNR}_{\text{curr}} = \frac{\eta \cdot N_\gamma}{\sqrt{\eta N_\gamma + N_d + N_r}}$ $\eta = \text{QE} \cdot absorb$

for large numbers $\text{SNR}_{\text{shot}} = \frac{\overline{n}}{\sqrt{\overline{n}}} = \sqrt{\overline{n}} \;\hat{=}\; \eta N_\gamma$

5.2 (Optical) Detectors

Photoel. effect $E_{\text{ph}} = hf = hc/\lambda = \phi + E_{\text{kin}}$

incident E_{ph} = E_{binding} + E_{kin} of ejected el.

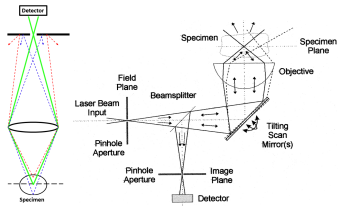
Photo mult. (PMT) $I = \alpha\,(S \cdot E_{\text{ph}}\,n/\Delta t)$, S: sensitivity, N_{d} ↑

Choose **binning mode** (2x2, 4x4, ...) such that the pixel size of the camera fits the best the maximum pixel size.

max. pixel size $\max = 0.3\frac{\lambda}{\text{NA}} \times M$

5.3 Imaging Deep Tissues

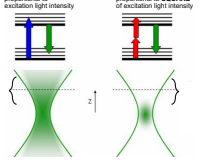
Beer-Lambert's law $I = I_0\,\text{e}^{-\mu x}$



1st Pinhole by illumination:
→ focus the illum. to a small spot
2nd Pinhole by detector:
→ reject out of focus light
→ collect light to PMT
Signal and Noise (**Pinhole size**):
- more out of focus light → blurry
- less in-focus light → SNR↓

Choose **pinhole size** $\sim\Delta r = 0.61\frac{\lambda}{\text{NA}} \times M$

5.3.1 2ph-Excitation



+ wavelength↑ → less absorbing/scattering
+ photons need to coincide in space and time to excite
+ No pinhole by detector for 2ph: Collect all the light
– low prob. of occurrence i.e. inefficient
– ultra-short pulse light source needed

6 Optical Biosensors

6.1 EM

Relations:

$c_0 = \frac{1}{\sqrt{\epsilon_0\mu_0}}$ $k_0 = \frac{2\pi}{\lambda_0} = \frac{\omega}{c_0} = \omega\sqrt{\epsilon_0\mu_0}$ $k = \frac{2\pi n}{\lambda}$ $n = \frac{c_0}{c} = \sqrt{\epsilon_r\mu_r} \simeq \sqrt{\epsilon_r}$

Dispersion relation $k^2 = \epsilon\omega^2 = \epsilon k_0^2 c_0^2 = k_0\frac{\epsilon}{\epsilon_0} = k_0^2\epsilon_r = (k_0 n)^2$

Plane waves $\vec{E}(\vec{r},t) = \vec{E}_0\,\text{e}^{j(\vec{k}\vec{r} - \omega t)}$

whereby $\partial_t \hat{=} -j\omega$, $\partial_x \hat{=} jk_x$, $\partial_z \hat{=} jk_z$, $\partial_y \hat{=} 0$ (infinite extent)

Depth of penetration $d_p = \frac{\lambda}{4\pi}\frac{1}{\sqrt{n_{\text{inc}}^2\sin^2\theta - n_2^2}}$ (about $500\,\text{nm}$)

6.2 Evanescent Field Techniques

Recap Always **total reflection** for $n_{\text{inc}} > n_2$

Evanescence Negligible change of sensitivity compared to the size of the antibodies

6.2.1 SPR – Surface Plasmon Resonance

Plasmon quantum of electron density wave in a metal

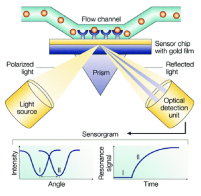
Plasmon Polariton mixt. of photon (diel.) and el. dens. wave (met.)

SPP field components point in dir. of propagation

Dispersion relation $k_{z,i}^2 = k_0^2\epsilon_i - \beta^2 \quad i = d, m$

Mom. of inc. wave $\beta := k_x = \frac{\omega}{c}\frac{\epsilon_m\epsilon_d}{\epsilon_m + \epsilon_d} = k_0 \cdot N$

where N effective refr. index of SP



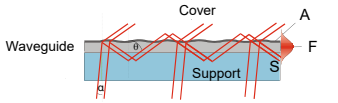
Launch a SPP: $0 \leqslant k_{||,\mathrm{inc}} \leqslant k_0\, n_{\mathrm{inc}}$

must ensure $\beta \overset{!}{=} k_{||,\mathrm{inc}} = k_0\, n_{\mathrm{inc}}\, \sin\theta \qquad \theta \in [0, \frac{\pi}{2}]$

$$\Longrightarrow \quad \boxed{n_{\mathrm{inc}} \geqslant n\sqrt{\frac{\epsilon_m}{\epsilon_m + n_c^2}}}$$

$$\boxed{\frac{\Delta\theta}{\Delta N} \hat{=} \frac{\partial\theta}{\partial N} = \frac{1}{n_{\mathrm{inc}} \cos\theta} \simeq \frac{1}{n_{\mathrm{inc}}}}$$

6.2.2 OWLS – Optical Waveguide Lightmode Spectroscopy



$N := n_F \sin\theta = n_{\mathrm{air}} \sin\alpha + \frac{l\lambda}{\Lambda}$

Penetr. depth $\boxed{\sigma \sim \frac{\lambda}{2\pi} \frac{1}{\sqrt{N^2 - n_c^2}}}$

Waves have to be in phase (constr. interf.) → extremely sensitive

constr. interference $0 = 2\pi m \overset{!}{=} \phi_F + \phi_{FS} + \phi_{FAC}$ ϕ: phase shifts

Ansatz for *3 layer model*

$$\begin{cases} \text{Cover:} & C\, \mathrm{e}^{-|k_{z,C}|\,(z - d_F/2)} \\ \text{Waveguide:} & B\, \mathrm{e}^{j k_{z,F} z} + A\, \mathrm{e}^{j k_{z,F} z} \\ \text{Support:} & D\, \mathrm{e}^{|k_{z,S}|\,(z + d_F/2)} \end{cases}$$

Idealized adlayer $n_A = n_C + c_A \frac{\mathrm{d}n}{\mathrm{d}c}$

Mass calculation $M = \mathrm{d}_A \frac{n_A - n_C}{\mathrm{d}n/\mathrm{d}c}$

6.3 Limitations

Above methods not usable for *diagnostic purpose* → NSB, LOD

Solution: diffractometric biosensors (Focal Molography)

7 Molecular Adsorption and Electron Transfer

7.1 Quantum Mechanics

Schrödinger eq. $\hat{H}\psi(\vec{r}) = E\psi(\vec{r})$ where $\hat{H} = -\frac{\hbar}{2m}\frac{\partial^2}{\partial x^2} + V$

1D potential well $\psi(x) = \sqrt{\frac{2}{L}} \sin(k_n x)$ for $0 \leqslant x \leqslant L$

where $k_n = \frac{n\pi}{L}, n \in \mathbb{N}$ and $E_n = \frac{\hbar^2 k_n^2}{2m}$

1D potential barrier (tunneling)

$$\psi(x) = \begin{cases} A\mathrm{e}^{jk'x} + B\mathrm{e}^{-jk'x} & x \leqslant 0 \\ C\mathrm{e}^{jk''x} + D\mathrm{e}^{-jk''x} & \text{for } 0 \leqslant x \leqslant d \\ F\mathrm{e}^{jk'x} & \text{otw.} \end{cases}$$

where $k' = \sqrt{2mE}/\hbar$ and $k'' = \sqrt{2m(V_0 - E)}/\hbar$

transm./tunneling probability $T = \frac{F^* F}{A^* A} = B\mathrm{e}^{-\beta d}$, $\beta = -2\sqrt{2m(V_0 - E)}/\hbar$

7.2 Electronic Transport through Molecules

Quantum conductance (1D) $G = G_0 \cdot T \quad G_0 = \frac{2e^2}{h}$, in parallel: $G = N \cdot G_0$

Tunneling probability $T = B\mathrm{e}^{-\beta d} \propto \mathrm{e}^{-\beta d} = \left(\mathrm{e}^{-\beta_{NPd}}\right)^N$

1D channel current $j = -(\mu_1 - \mu_2) e\, v\, \rho_E \qquad \rho_E = DOS = \frac{1}{\hbar\pi}\sqrt{2m/E}$

7.3 Atomic and Molecular Orbitals

Bond order: defined by difference (#electrons) divided by two

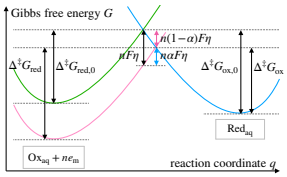
→ If the bond order is different from zero, then the bond is *stable*

7.4 Transition State Theory $A + B \rightarrow [AB] \rightarrow C + D$

Gibbs free energy $G = H - TS = U + pV - TS$

Ideal gas law $pV = n_{\mathrm{total}} RT$

7.4.1 Marcus Theory



See **section ??** (Butler-Volmer)

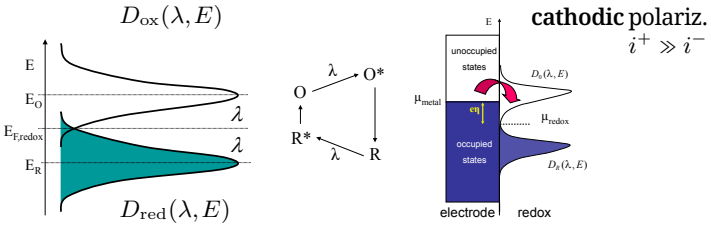
$$k_{\mathrm{red}} = v_{\mathrm{red}}\, \mathrm{e}^{-\frac{\Delta^\ddagger G_{\mathrm{red}}}{RT}}$$

$$k_{\mathrm{ox}} = v_{\mathrm{ox}}\, \mathrm{e}^{-\frac{\Delta^\ddagger G_{\mathrm{ox}}}{RT}}$$

$$\Delta^\ddagger G_{\mathrm{red}} = {}^\ddagger G - G_{\mathrm{ox,min}}$$

$$\Delta^\ddagger G_{\mathrm{ox}} = {}^\ddagger G - G_{\mathrm{red,min}}$$

7.4.2 Gerischer's view



cathodic polariz. $i^+ \gg i^-$

8 Potentiometric Biosensors

8.1 Redox Reaction

oxidation: e^- donor (*anode*) **reduction**: e^- acceptor (*cathode*)

Chemical reaction $\mathrm{A \cdot ox + B \cdot X + z \cdot e^- \underset{k_{ox}}{\overset{k_{red}}{\rightleftharpoons}} C \cdot red + D \cdot Y}$

reaction quotient $Q = \frac{a_{ox}^A \cdot a_X^B}{a_{red}^C \cdot a_Y^D}$ • $a_{\mathrm{ion}} = r_{\mathrm{ion}}[\mathrm{ion}]/1\,\mathrm{mol/l}$ • $a_{\mathrm{solids}} =$ • $[\mathrm{H_2O}] =$

Standard conditions "STP" $T = 1\,°\mathrm{C}, \quad p = 101.3\,\mathrm{kPa} = 1.013\,\mathrm{bar}$

$[\mathrm{ion}] \equiv c_{\mathrm{ion}} = 1\,\mathrm{mol/l}, \quad r = 1$ "activity coeff."

8.2 Electrochemistry

Dynamic equi. no net charge over time ($k_1 = k_{-1}$)

Chemical potential $\mu_j = \left(\frac{\partial G}{\partial n_j}\right)_{p,T,n'}$

Equilibrium $\mu_A = \mu_B$ general: $\sum_{\mathrm{prod}} \nu_j \mu_j = \sum_{\mathrm{react}} \nu_j \mu_j$

Chemical potential $\mu_{\mathrm{redox}} \equiv E_{F,\mathrm{redox}} \equiv E^*$

Equilibrium $\bar{\mu}_A = \bar{\mu}_B \qquad \bar{\mu}_j = \mu_j^0 + z_j F \Delta\phi$

Contact potential

$$\Delta\phi = V_{\mathrm{in}} - V_{\mathrm{out}} = \frac{k_B T}{Q} \ln\left(\frac{[C]_{\mathrm{out}}}{[C]_{\mathrm{in}}}\right)$$

$$= -\frac{\Delta_r G}{zF} = \frac{1}{zF}\left(\sum_{\mathrm{ox}} \nu_j \mu_j - \sum_{\mathrm{red}} \nu_j \mu_j + z\mu_e\right)$$

Nernst eq., $\Delta\phi$ $E = E^0 - \frac{59\,\mathrm{mV}}{z} \log_{10} Q$ $\frac{RT}{zF} \ln Q = \frac{2.303RT}{zF} \log_{10} Q$

$E_{\mathrm{cell}} = E_1 - E_2$ or let $E_1 \overset{!}{=} E_2 \rightarrow \mathrm{pH} = \ldots$

8.3 Ion-selective Electrodes (ISE)

Sensor is seperated to solution through a H^+ permeable glass.

pH electrode $\mathrm{pH} = -\log_{10} a_{\mathrm{H^+}} = \frac{K' - \Delta\phi}{0.059\,\mathrm{V}}$

8.4 Bioenzymatic Electrodes

Find out how much of an enzyme was initially present

$\boxed{\mathrm{substrate + X + \ldots \xrightarrow{enzyme} product + Y + \ldots}}$

Detect **"a lot"** X

↔ **"very few"** substrate was initially present

9 Amperometric Sensors

9.1 Electrochemistry

Overpotential $\eta \equiv \Delta\phi_{\mathrm{appl}} - \Delta\phi^0 \equiv \phi_s - \phi_m \equiv E - E^0$ E^0: Nernst

$\eta = 0$ **(equilibrium** i.e. no net current)

Faraday's law of electrolysis *1st:* $n \propto Q$, *2nd:* (equiv. weigth) $W_{\mathrm{eq}} = M/z$

$m = \frac{Q}{z\,F}\, M = \frac{I\,t}{z\,F}\, M = n \cdot M$ $[m]$=g, $[n]$=mol

9.1.1 Butler-Volmer equation Effect of η on barrier heigth G

Let ϕ_s: potential of ions in solution

ϕ_m: potential of e^- in (metallic) electrode

z: valency of oxidized species

n: #of transferred e^-

$z - n$: valency of reduced species

transfer coeff. $\alpha \qquad nF\eta = n(1-\alpha)F\eta + n\alpha F\eta$

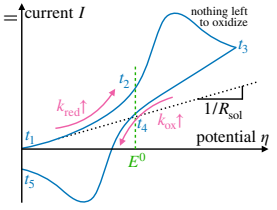
@equi. ($\eta = 0$)

$\boxed{f \equiv \frac{F}{RT} \overset{@298\,\mathrm{K}}{=} \frac{1}{25.69\,\mathrm{mV}}}$ $k_{\mathrm{red}} = k_0\, \mathrm{e}^{-n\alpha f\eta}, \quad k_{\mathrm{ox}} = k_0\, \mathrm{e}^{n(1-\alpha)f\eta}$ $k_0 \equiv k_{\mathrm{red}} = k_{\mathrm{ox}}$

The Butler-Vomer eq. relates what we measure (the **current**) with what we would like to determine (the **concentration** of an analyte):

Butler-Volmer $j = \underbrace{nFAk_0}_{j_0}\left(C_{\mathrm{ox}}(0,t)\, \mathrm{e}^{-\alpha n f\eta} - C_{\mathrm{red}}(0,t)\, \mathrm{e}^{(1-\alpha)nf\eta}\right)$

9.2 Cyclic Voltammetry



Offers information on the mechanism of the ec reactions occuring at an electrode.

- May be irreversible
  → #upper peaks ≠ #lower peaks
  → irreversible reaction fast
- Sweep rate
  → irreversible reaction slow

Water electrolysis High potentials/volt's
→ affects I (disturbance, bad)

9.3 Amperometric Sensors

9.3.1 Clark (Oxygen) Electrode → Measure O_2

- test solution $\xrightarrow{O_2 \text{ passes membrane}}$ Pt cathode $\xrightarrow{\text{reduction}}$ current
- Ag anode is in KCl solution (→ "enough" Cl^- for oxidation)

9.3.2 1st and 2nd Generation

analyte of interest ... $\underbrace{\text{redox enzyme}}_{\text{catalize}}$ $\underbrace{\ldots}_{\text{P1}}$ $\underbrace{\text{mediator}}_{\text{P2}}$ ... Clark Electrode

problem P1 (1st) O_2 may be consumed, when not measured

problem P2 (2nd) need to have "enough" of them

Mediator has to be • reversible • not toxic • no side reactions

9.3.3 3rd Generation

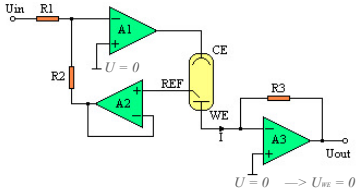
Immobilization/fixation of a redox enzyme on electrode surface

→ free-diffusing redox mediators are not necessary

→ in vivo measurments allowed since immobilized

problem P3 Efficient electron transfer (Marcus theory)
→ may be overcome by mediators
→ minimize ET distance

9.4 Three Electrode Cell



$$\frac{U_{REF}}{R_2} = -\frac{U_{in}}{R_1}$$
$$I = \frac{U_{out}}{R_3}$$

10 Membranes and Transport

10.1 Diffusion

Concentration $c(x,t)$ $[\mathrm{mol/l} = \mathrm{mol/10^3\,cm^3}]$

Flux $\phi(x,t)$ $[\mathrm{mol/cm^2\,s}]$

Fick's law $\phi(x,t) = -D_n \frac{\partial c_n(x,t)}{\partial x} = \phi_0$ in **steady state**

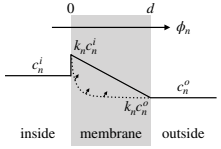
D_n: diffusion coeff $[\mathrm{cm^2/s}]$

Continuity eq. $-\frac{\partial \phi_n(x,t)}{\partial x} = \frac{\partial c_n(x,t)}{\partial t}$

e.g. **Delta initial condition** $c(x,t)|_{t=0} = n_0\delta(x)$

$c(x,t) = \frac{n_0}{\sqrt{2\pi\sigma}} e^{-x^2/\sigma^2}$, $\sigma = \sqrt{2D_n t}$, $t_{1/2} = \frac{1}{D_n}x_{1/2}^2$

10.1.1 Diffusion across membrane



$\phi_n(t) = P_n\,(c_n^i(t) - c_n^o(t))$

with permeability: $P_n = \frac{D_n k_n}{d}$

state space (inside membrane)

after $\tau_{SS} = \frac{d^2}{\pi^2 D_n}$

10.2 Osmosis

Osmotic (back-)pressure $\pi(x,t) = RT\sum_n c_n(x,t)$ $c_\Sigma(x,t)$: osmolarity/ total conc.

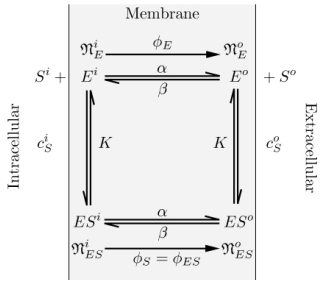
Zero pressure if $\pi_i - \pi_o = 0$

Volume of the cell $v_c(\infty) = v_c' + \frac{N_\Sigma^i}{c_\Sigma^o}$ v_c': non-water volume

10.3 Carrier mediated Transport

Hints for their existance
- Saturation of solute trsp. (↯ Fick's law)
- Competitive inhib.
- Structure specificy

4-state carrier model



1. Conservation
   $N_E^i + N_E^o + N_{ES}^i + N_{ES}^o = N_{ET}$
2. Fast binding (always steady state)
   $K = \frac{c_S^i N_E^i}{N_{ES}^i} = \frac{c_S^o N_E^o}{N_{ES}^o}$
3. Translocation (char. by flux)
   $\phi_{E(S)} = \alpha N_{E(S)}^i - \beta N_{E(S)}^o$
4. Zero net flux
   $\phi_E + \underbrace{\phi_{ES}}_{\phi_S} = 0$

Net flux out of cell $\phi_S = \frac{\alpha\beta}{\alpha+\beta} N_{ET}\left(\frac{c_s^i}{c_s^i+K} - \frac{c_s^o}{c_s^o+K}\right)$ $(\phi_S)_{max} = \frac{\alpha\beta}{\alpha+\beta}N_{ET}$

10.4 Ion Transport

Continuity eq. $\frac{\partial J_n(x,t)}{\partial x} = -z_n F \frac{\partial c_n(x,t)}{\partial t}$ z_n: valency

Poisson eq. $\frac{\partial^2 \psi(x,t)}{\partial x^2} = -\frac{1}{\epsilon}\sum_n z_n F c_n(x,t)$

i.e. electric field potential = local charge density

Nernst-Planck eq. u_n: molar mobility

$$\underbrace{J_n(x,t)}_{\text{current}} = \underbrace{-z_n F D_n \frac{\partial c_n(x,t)}{\partial x}}_{\text{diffusion of charge}} \underbrace{- u_n z_n^2 F^2 c_n(x,t)\frac{\partial \psi(x,t)}{\partial x}}_{\text{mobility of charge in E-field}}$$

Debye Length "depletion width": $\lambda_D \simeq 1\,\mathrm{nm}$ within $\tau_r \simeq 1\,\mathrm{ns}$

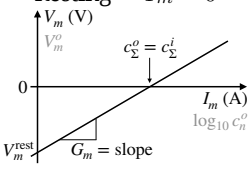
Nernst Equi. Pot. $V_n = \frac{59\,\mathrm{mV}}{z_n}\log_{10} Q$ with $Q = \frac{c_s^o}{c_s^i}$, $\frac{D_n}{u_n} = RT$

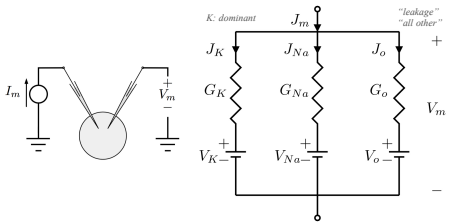
$V_K \simeq -75\,\mathrm{mV}$, $V_{Na} \simeq 55\,\mathrm{mV}$

Donnan equil. $V_m = V_n \Rightarrow I_n = 0$ $\left(\frac{c_n^o}{c_n^i}\right)^{\pm\frac{1}{z_n}} = e^{\frac{FV_m}{RT}}$, +: anodic

10.4.1 Resting (membrane) potential

Resting $\leftrightarrow I_m = 0$





3-ion model permeable to K, Na and some others

Resting potential $V_m^{rest} = \frac{V_K G_K}{G_{tot}} + \frac{V_{Na} G_{Na}}{G_{tot}} + \frac{V_o G_o}{G_{tot}}$ $G_{tot} \equiv G_m = \text{slope}$

$J_m = J_K + J_{Na} + J_o = 0$ where $J_n \neq 0$ (only momentarly rest)

10.5 Active Transport

Rest is only momentarly → maintain conc. grad with *act. transport*
- Ions moving against its conc. gradient → ATP needed
- Active pumps for $J_m = 0$ (rest) and $J_{n,in} + J_{n,out} = 0$ (quasi equilib.)

11 Action Potential & Hodgkin-Huxley Model

11.1 Current Clamp

Fix current I_m and measure membrane potential

→Good for observing AP (since voltage can change)

11.2 Voltage Clamp

Fix voltage V_m and measure the current

→Good for studying membrane channel proteins (ion transp.)

11.3 2-state ion channel model

two states close $\underset{\beta}{\overset{\alpha}{\rightleftharpoons}}$ open with gate charge Q

#open states $\frac{dn(t)}{dt} = \alpha(\mathcal{N} - n(t)) - \beta n(t)$

Prob. being open $x(t) \simeq \frac{n(t)}{\mathcal{N}} = x_\infty + (x_0 - x_\infty)\,e^{-t/\tau_x}$

$x_\infty = \frac{\alpha}{\alpha+\beta}$, $\tau_x = \frac{1}{\alpha+\beta}$

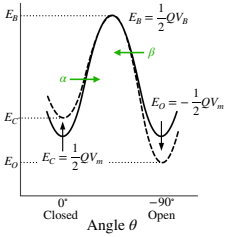
Boltzmann law

$\alpha = A\,e^{(E_c - E_B)/kT}$

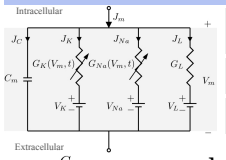
$\beta = A\,e^{(E_o - E_B)/kT}$

$x_\infty = \frac{1}{1+\beta/\alpha} = \frac{1}{1+e^{-QV_m}}$

$\tau_x = \frac{1}{A\,e^{-\frac{1}{2}QV_B/kT}} \cdots \frac{1}{e^{\frac{1}{2}QV_m/kT} + e^{-\frac{1}{2}Q}}$



11.4 Circuit Model (Conductance) $V_K \simeq -75\,\mathrm{mV}$, $V_{Na} \simeq +55\,\mathrm{mV}$

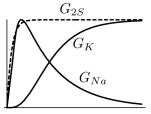


$J_C = C_m \frac{dV_m(t)}{dt}$

$J_m = C_m \frac{dV_m}{dt} + G_m(V_m - V_m^{rest})$

$J_K = G_K\,(V_m - V_K)$ always > 0, we don't fall below ($V_m > V_K$)

$J_{Na} = G_{Na}\,(V_m - V_{Na})$ $\begin{cases} > 0 & \text{if } V_m - V_{Na} > 0 \\ < 0 & \text{otw.} \end{cases}$



- slow onset; decays slow
- K^+ inactivates Na^+ channels
- → **Refractoriness** (kurzzeitige AP Resistenz) (Na^+ channels are still inactivate, $h \sim 0$)

11.5 Multiple states gate

open $n_Q(V_m, t)$ big/small Q → fast/slow dynamics

Let's write $\boxed{h} = n_{-Q}$ "slow", $\boxed{m} = n_{2Q}$ "fast", $\boxed{n} = n_Q$

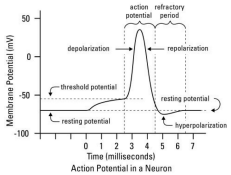
The *lower* T, the *bigger* the diff. between them.

11.6 Hodkin-Huxley (HH) model

Experimentally you may fit the K^+ and Na^+ current into

Conductance $G_K(V_m,t) = \overline{G}_K\,n^4$

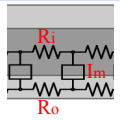
$G_{Na}(V_m,t) = \overline{G}_{Na}\,m^3 h$



Four regimes during AP:
II *De*polarization: Na^+ gate opens → Na^+ in
III *Re*polarization: K^+ gate opens → K^+ out
IV *Hyper*polarization:
Active $Na^+ - K^+$ – ion pump ($Na^+ \leftrightarrow K^+$)

11.7 Decrement-free conduction

11.7.1 Core – Conductor Model



$R_i = r_i\,dz$ ← dz: per unit length

$R_o = r_o\,dz$

$I_m = k_m\,dz$ ← may use HH-model

Core–Conductor eq. $\frac{\partial^2 V_m(z,t)}{\partial z^2} = (r_o + r_i)K_m(z,t) - \overbrace{r_o K_e(z,t)}^{\overset{\text{if}}{=} 0 \to I_o = -I_i}$

wave eq. $= \frac{1}{v^2}\frac{\partial^2 V_m(z,t)}{\partial t^2}$ with $v = \frac{W}{\Delta t}$

$\overset{K_e = 0}{\Longrightarrow}$ $v \simeq \frac{K_m a}{2\rho_i}$ for $r_i \gg r_o$ i.e. $v \propto \sqrt{a}$

11.7.2 Cable model – Core–Conductor with HH-model inside

Linearize (1st order) timescale for membrane voltage changes $\tau_m = \frac{C_m}{G_m}$

Cable equation Let $v_m = V_m + V_m^{rest}$

$v_m + \underbrace{\tau_m \frac{\partial v_m}{\partial t}}_{=0 \text{ time indep.}} - \lambda_C^2 \frac{\partial^2 v_m}{\partial z^2} = r_o \lambda_C^2 K_e$ $\lambda_C = \frac{1}{\sqrt{g_m(r_o + r_i)}}$

11.7.3 Saltatory Conduction Hypothesis

→ explain discrete manner in steps

velocity ~ axon diameter D total delay $\sim N(\#\text{nodes}) \sim (\text{axon length})/L$

$\overset{L \sim D}{\longrightarrow}$ velocity $\sim \frac{\text{total delay}}{\text{axon length}} \sim D$

